Band-1 Hadamard Order 3716 — Miyamoto (Double-seed H_928); Gap 2052 Still OPEN

H_464/H_928/C_930/H_3716 PASS; 3716 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3524_Miyamoto`; `2026-09-15_Band1_Hadamard_3044_Miyamoto` (Double-seed pattern).
Lab. `/workspace/experiments/band1/hadamard_3716_miyamoto/`.
Cutoff. 2026-09-15T12:48:14Z (~8:48AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 464, 928, and 3716 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double-seed $H_{928} = \text{Sylvester-double}(H_{464})$ where H_{464} is Paley type I ($q = 463$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3716 is a **Miyamoto-only bonus** (NOT in SPK): Paley-I **FAIL** ($q = 3715$ not prime); Paley-II **FAIL** ($q = 1857$ not prime); Kronecker **FAIL** ($3716 = 4 \cdot 929$, 929 not Hadamard-admissible). Seed path: Paley-I for 928 **FAIL** ($q = 927$ not prime); Paley-II for 928 **FAIL** ($q = 463 \equiv 3 \pmod 4$). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3524 on box (Kronecker seed). This increment ships **order 3716** ($3716 = 4 \cdot 929$, 929 prime $\equiv 1 \pmod 4$) via the **Double-seed** pattern proven at 3044. Seed H_{464} via Paley type I ($q = 463$); H_{928} = Sylvester-double(H_{464}); symmetric Paley conference C_{930} over prime 929; H_{3716} = Miyamoto(H_{928}, C_{930}). Unit tests **28/28 PASS**. SPK next gap remains **2052** (OPEN). Order 3716 is **Miyamoto-only** (NOT in SPK). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3716; Miyamoto; Sylvester doubling; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{464} constructed and verified (Paley type I, $q = 463$). 2. H_{928} constructed and verified (Sylvester double of H_{464}). Paley-I/II seeds for 928 **rejected**. 3. C_{930} constructed and verified (Paley conference, $q = 929$): $CC^\top = 929I$, zero diagonal. 4. H_{3716} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $3716 \notin$ SPK (Miyamoto-only bonus). Paley-I/II/Kronecker paths for 3716 **rejected**. Next constructive gap remains **2052** (OPEN). Orders 464 and 928 **are** classical SPK-eligible (Paley-I / Sylvester double).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3716 = 4 \cdot 929$
$q = 929$	prime, $929 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{928}
Needed conference	order $930 = q + 1$
H_{464} path	Paley type I, $q = 463$ (prime $\equiv 3 \pmod 4$)
H_{928} path (used)	Sylvester double of H_{464}
H_{928} Paley-I	**REJECTED**: $q' = 927$ not prime
H_{928} Paley-II	**REJECTED**: $2(q'+1) = 928 \Rightarrow q' = 463$ prime but $\equiv 3 \pmod 4$
Conference path	Symmetric Paley over prime 929
Paley-I for 3716	**REJECTED**: $q' = 3715$ not prime
Paley-II for 3716	**REJECTED**: $q' = 1857$ not prime
Kronecker for 3716	**REJECTED**: only 4×929; $929 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{464}	Paley type I, $q = 463$	0	0.1509	**PASS**
H_{928}	Sylvester double of H_{464}	0	0.3712	**PASS**
C_{930}	Paley conference, $q = 929$	0	0.7795	**PASS**
H_{3716}	Miyamoto(H_{928}, C_{930})	0	22.1645	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 24.2117 s. Compressed ± 1 matrices saved as `H464_pm1.npz`, `H928_pm1.npz`, `C930_pm1.npz`, `H3716_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior ships (through 3524/880/…; **exclude** 3524 Miyamoto-only) $\cup$ verified H_{464}, H_{928} $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3716 does **not** enter SPK.

Quantity	Value
Known count	886
3716 in SPK	**False** (Miyamoto-only bonus)
928 in SPK	**True** (Sylvester double of Paley-I)
464 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**Yes** (like 3524/3284/3076/3044)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

28/28 PASS (Sylvester/Paley/Miyamoto controls; $q = 463/929$ arithmetic; Double path for H_{928}; Paley-II seed path for 928 rejected; SPK path rejection for 3716; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 3716 is **Miyamoto-only** (NOT folded into SPK).

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 937 \to n = 3748$ or next seedable) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_3716_miyamoto/` · *Script:* `verify_hadamard_3716.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.